CSE 4615:: Wireless Networks

Understanding Wireless Communication: Needs, Barriers, and Design Adaptations

Lecture 1

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What You Learn in This Course

- How does a Wireless Network Work?

- What are the Limitations of Wireless Communications?

- How to Address These Limitations to Boost Communication Performance?

Structure of the Course

- Wireless Access Networks
- Wireless Communication Fundamental
  - Radio Transmission Fundamentals
  - Radio Spectrum Regulation
  - Communication Requirements
- Mesh Networking Basic
- Wireless LAN (IEEE 802.11)
- VANET (IEEE 802.11p)
- Personal Area Network (IEEE 802.15)
- Wireless Mobile Network (5g , LTE)
- High Throughput (HT) Networks (IEEE 802.11n, MIMO)
- Advanced Techniques for Wireless Communication
  - Semantic Communication
  - ML & AI in Wireless Communications

Recourses of the Course

- Classroom Code: **3tj4qv4w**
- Invitation Link: https://classroom.google.com/c/Nzc3NjUyNjIwNDE1?cjc=3tj4qv4w

Book Title	Authors
IEEE 802 Wireless Systems, 1st Edition (Wiley)	B. H. Walke, S. Mangold and L. Berlemann
Wireless Communication Networks and Systems by, 1stEd	Beard and Stallings
Wireless Communications, 2nd Edition	Andrea Goldsmith
Wireless Communications and Networks, 2nd Edition	William Stallings

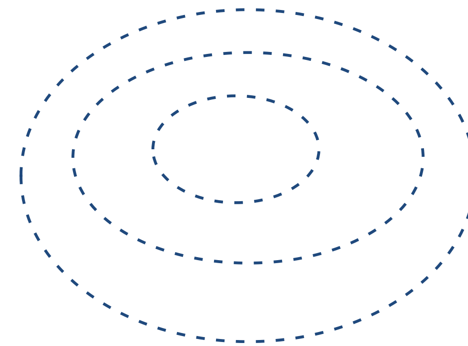
Understanding Wireless Communication:

Fundamentals of Communications

- Objective: move a **message** from point A to point B with the **required quality** at a **minimum cost** with respect to certain design criteria
- **Message:** typically, a number of packets or bits
- **Quality measures**
  - error probability, e.g., bit or packet or message error rate
  - delay: time from when information is available at transmitter until it is output from the receiver
  - data rate: number of transmitted bits/second
- **Resources and costs**
  - **power** of transmitted signal (power efficiency)
  - **bandwidth** of transmitted signal (spectral efficiency)
  - **complexity** of transmitter and receiver (hardware and software)

How do wireless and wired networks differ?

Wired Vs. Wireless Communication



Wired	Wireless
Each cable is a different channel	One media (cable) shared by all
Signal attenuation is low	High signal attenuation
Low interference (Cross-talk)	High interference noise; co-channel interference; adjacent channel interference

Wired Vs. Wireless Communication

Parameter	Wired Network	Wireless Network
Communication Medium	Copper, Fiber optic	Air (Radio Waves)
Standard	IEEE 802.3 (Ethernet)	IEEE 802.11 family (Wi-Fi)
Mobility & Roaming	Limited	High
Security ?	High (less vulnerable to attacks)	Lower (prone to eavesdropping/hacking)
Speed / Bandwidth		

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Reliability		
Flexibility to Change		

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Interference / Fluctuations		
Installation Activity		

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Installation Activity	Cumbersome, labor intensive	Easy, minimal labor
Installation Time		

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Maintenance Cost ?	High	Low
Related Equipment ?		
Key Benefits		install

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How to Address These Limitations of Wireless Networks to Boost Communication Performance?

What Are the Limitations of Wireless Communications?

Limitation	Description
Susceptible to Interference	From other wireless devices or physical obstacles (walls, appliances).
Signal Attenuation & Fading	Signals weaken over distance and vary due to multipath propagation.
Bandwidth Limitations	Shared medium leads to congestion and reduced throughput.
Mobility-Induced Handover Delays	Switching between access points can cause latency or packet loss.
Power Constraints	Battery-operated devices must optimize transmission to save power.
Security Vulnerabilities	Easier to intercept or attack (e.g., spoofing, eavesdropping).

Embracing Wireless: Key Motivations

Advantages

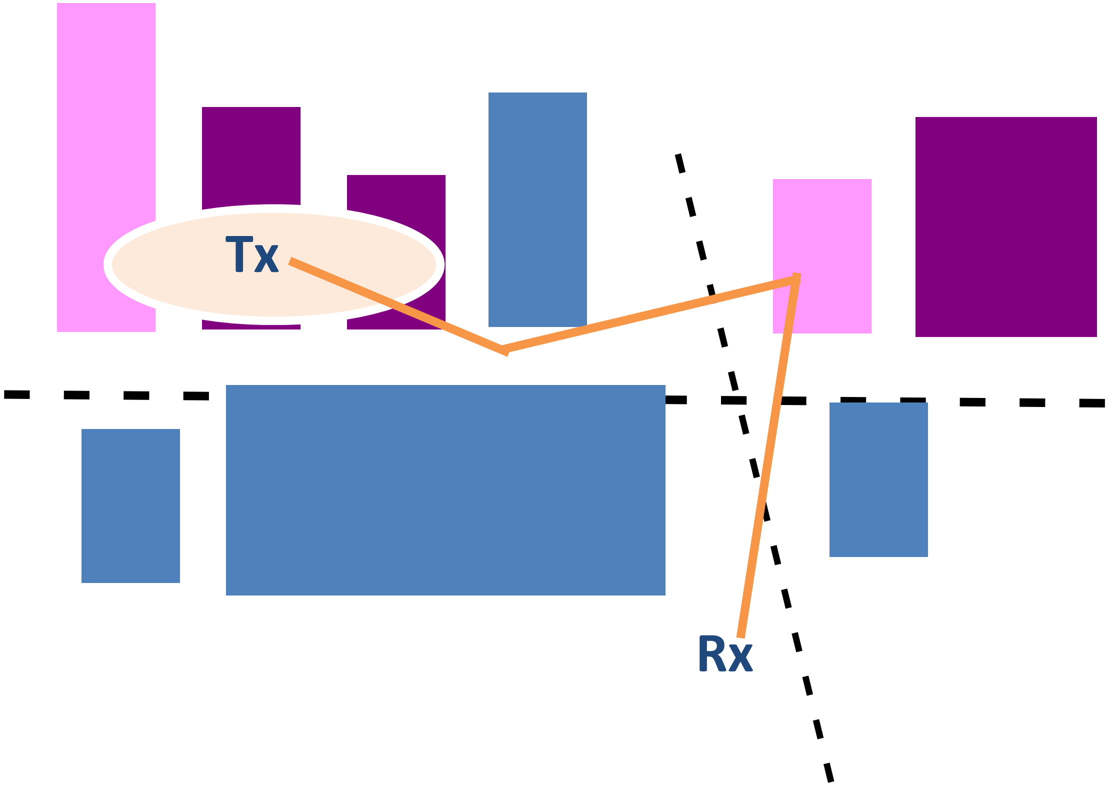
- Eliminates the need for physical cabling in difficult or remote areas
- Enables **user mobility** and flexible access
- Lower installation cost in many environments

How to Address These Limitations to Boost Communication Performance?

Challenge	Solutions
Interference	Use directional antennas, adaptive frequency selection, and smart AP placement.
Signal Attenuation	Deploy MIMO, repeaters, and mesh networks for better coverage.
Bandwidth Sharing	Cognitive Radio for Spectrum Sharing, Apply QoS (Quality of Service)
Handover Delays	Use fast handoff techniques and mobile IP support for seamless mobility.
Power Efficiency	Use energy-aware MAC protocols and sleep cycles.
Security Threats	Implement WPA3, VPNs, firewalls, and regular firmware updates.

Wireless Wireless links characteristics

Understanding Wireless Links



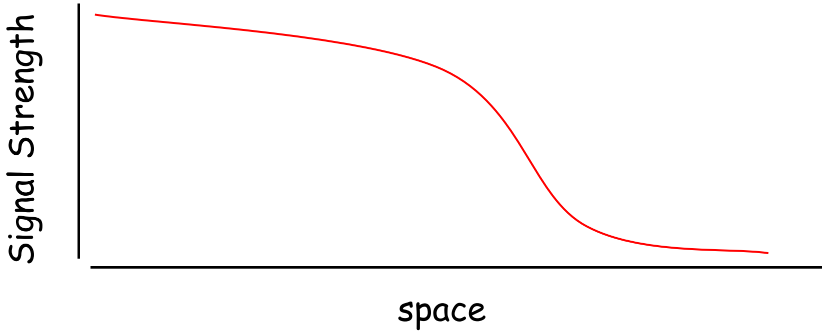
- How does signal propagate ?
- How much attenuation take place ?
- How does signal look like at the receiver ?

Wireless Link Characteristics

Wireless is a broadcast medium!

Differences from wired link ….

- decreased signal strength: radio signal attenuates as it propagates through matter (path loss)



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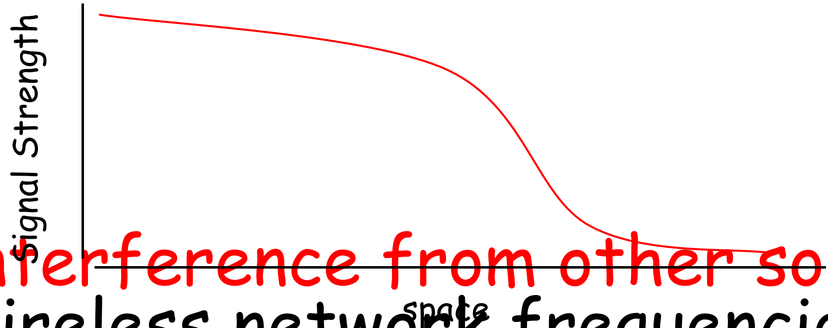


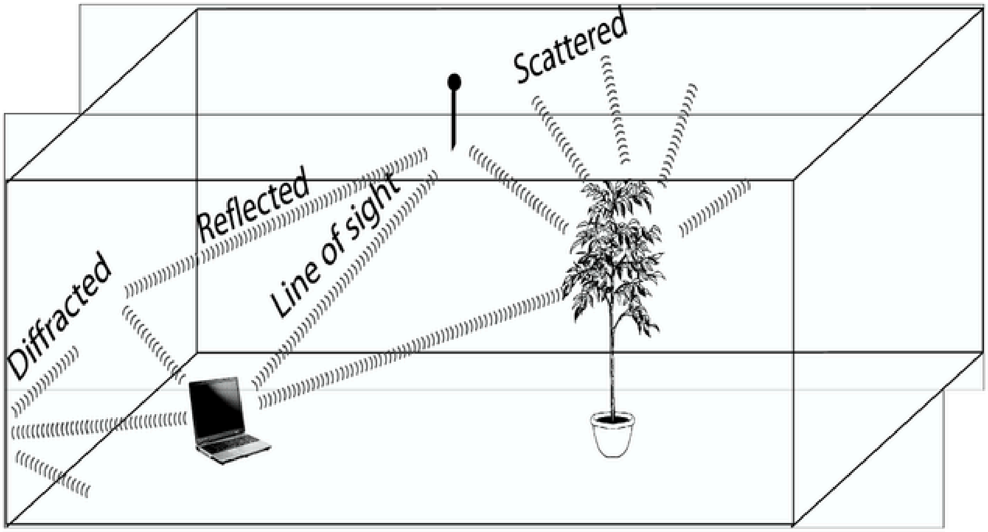
- interference from other sources: standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors, microwaves) interfere as well

Wireless Link Characteristics

Wireless is a broadcast medium!

Differences from wired link ….

- multipath propagation (fading): radio signal reflects off objects or ground, arriving at destination at slightly different times
  - Reflection
  - Diffraction
  - Scattering



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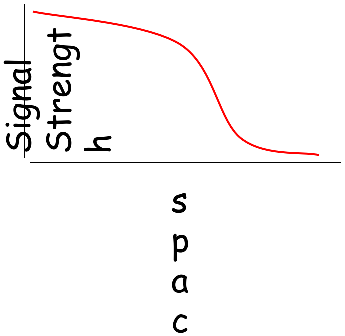
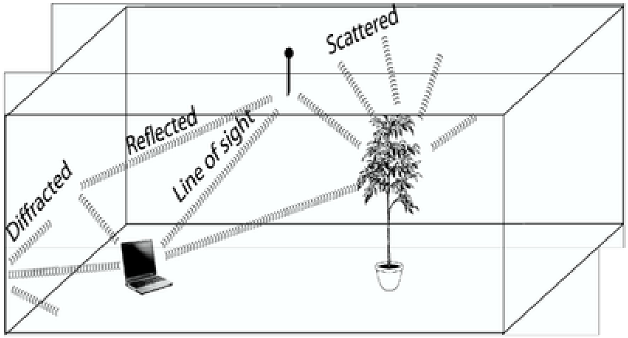
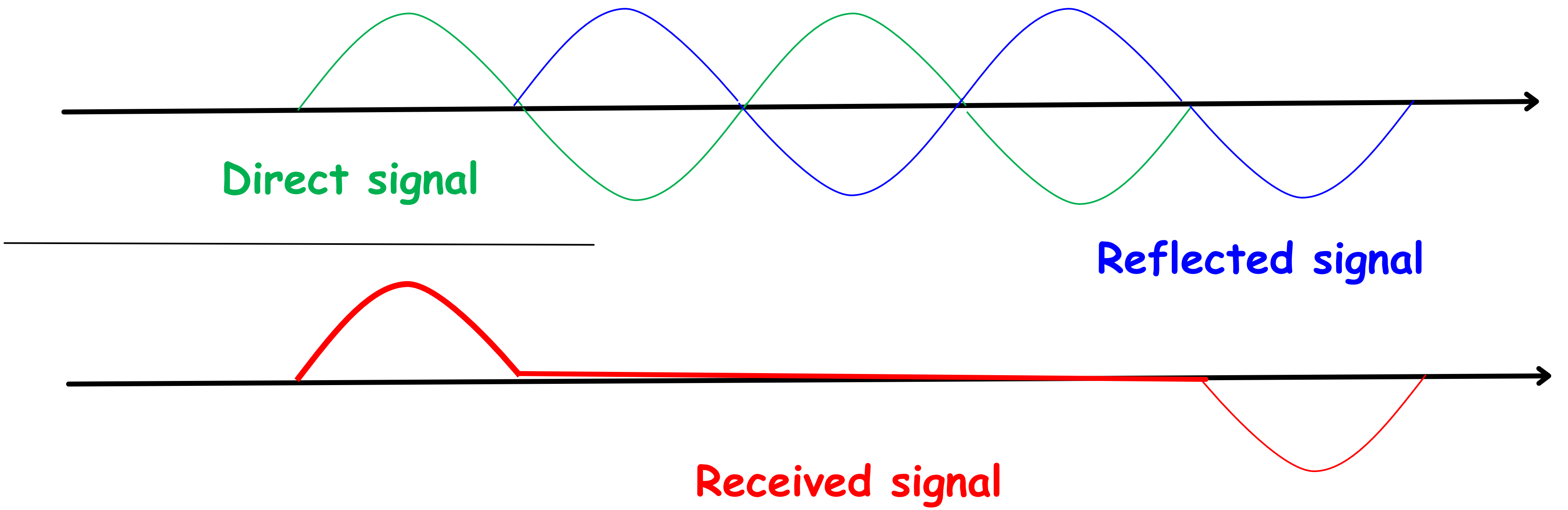




…. make communication across (even a point to point) wireless link much more "difficult"

Impact of Multipath Propagation



Wireless Link Characteristics- Fading

- Channel characteristics change over time and location
  - e.g., movement of sender, receiver and/or scatters
- quick changes in the power received (short term/fast fading)
- slow changes in the average power received (long term/slow fading)

power

long term fading

short term fading

t

Wireless Link Characteristics- Fading

Sample Example

10

10^{1}W

Path loss in decibel, $dB = 10 \log \left(\frac{P_1}{P_2} \right)$

Where:

- P_1 = Transmitted Power
- P_2 = Received Power

Power

Path loss from source to d2 = 70dB

10^{-3}

1 mW

10,000 times

40 dB

1,000 times

30 dB

1 □W

10^{-6}

sourc e

d1

d2

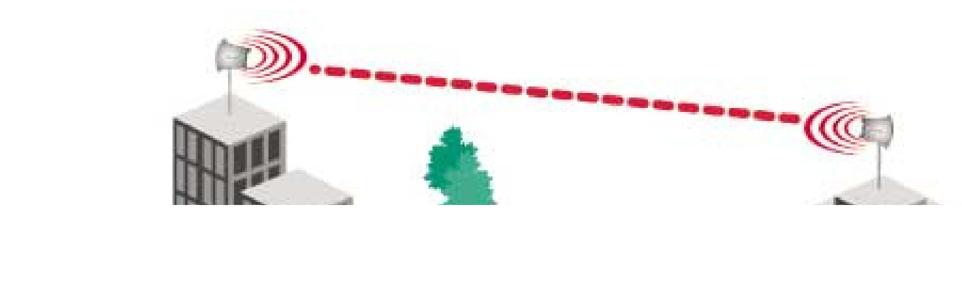
Wireless Link Characteristics- Fading

To convert power from milliwatts (mW) to decibel-milliwatts (dBm), use the formula:

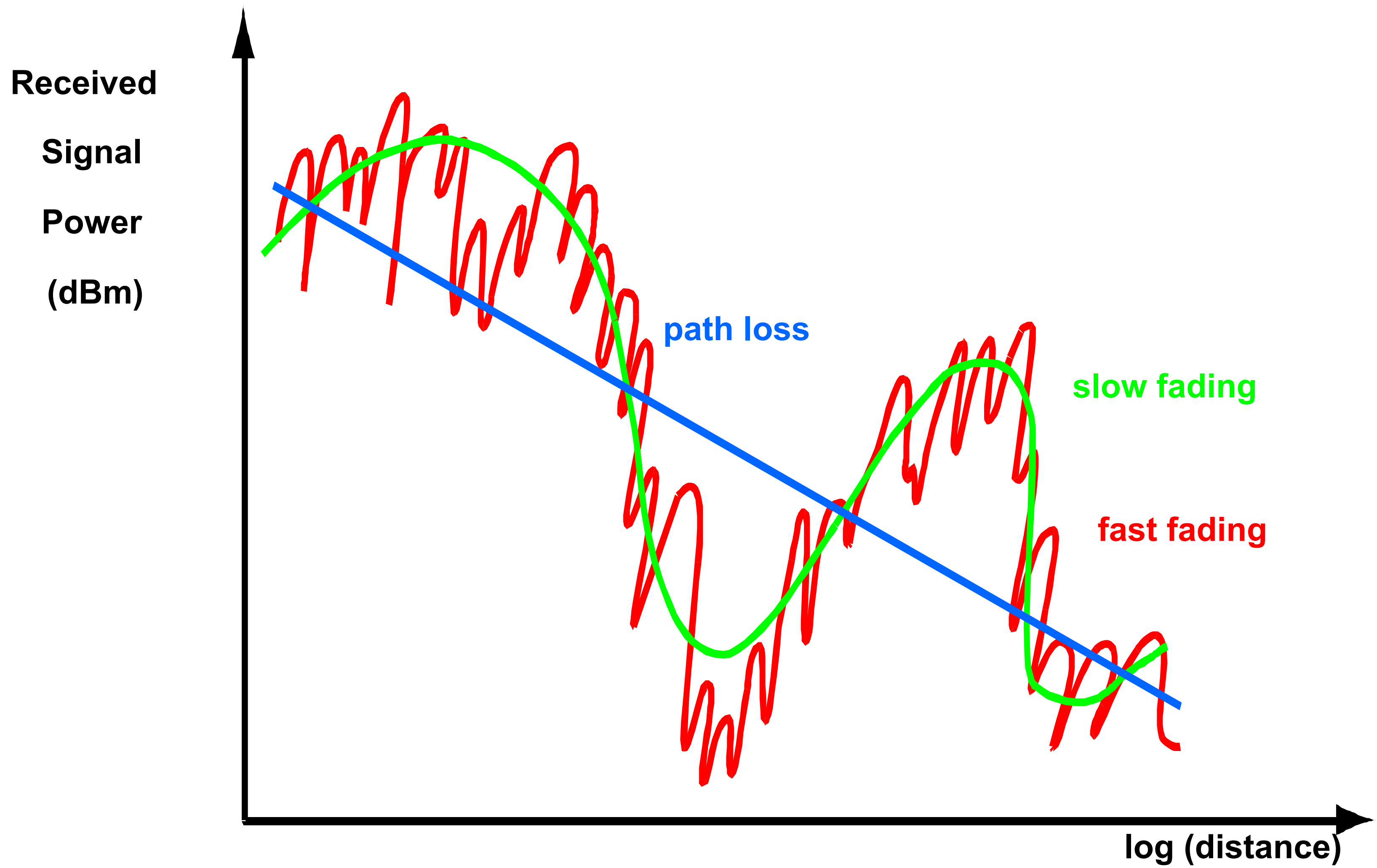
$$P\ (dBm) = 10 \cdot \log10(P\ (mW))$$

If,
Transmitted Signal Power: Ptx decibel-milliwatts (dBm)
Receiver Sensitivity: Prx decibel-milliwatts (dBm)

Maximum Tolerable Path Loss = Ptx (dBm) – Prx (dBm)



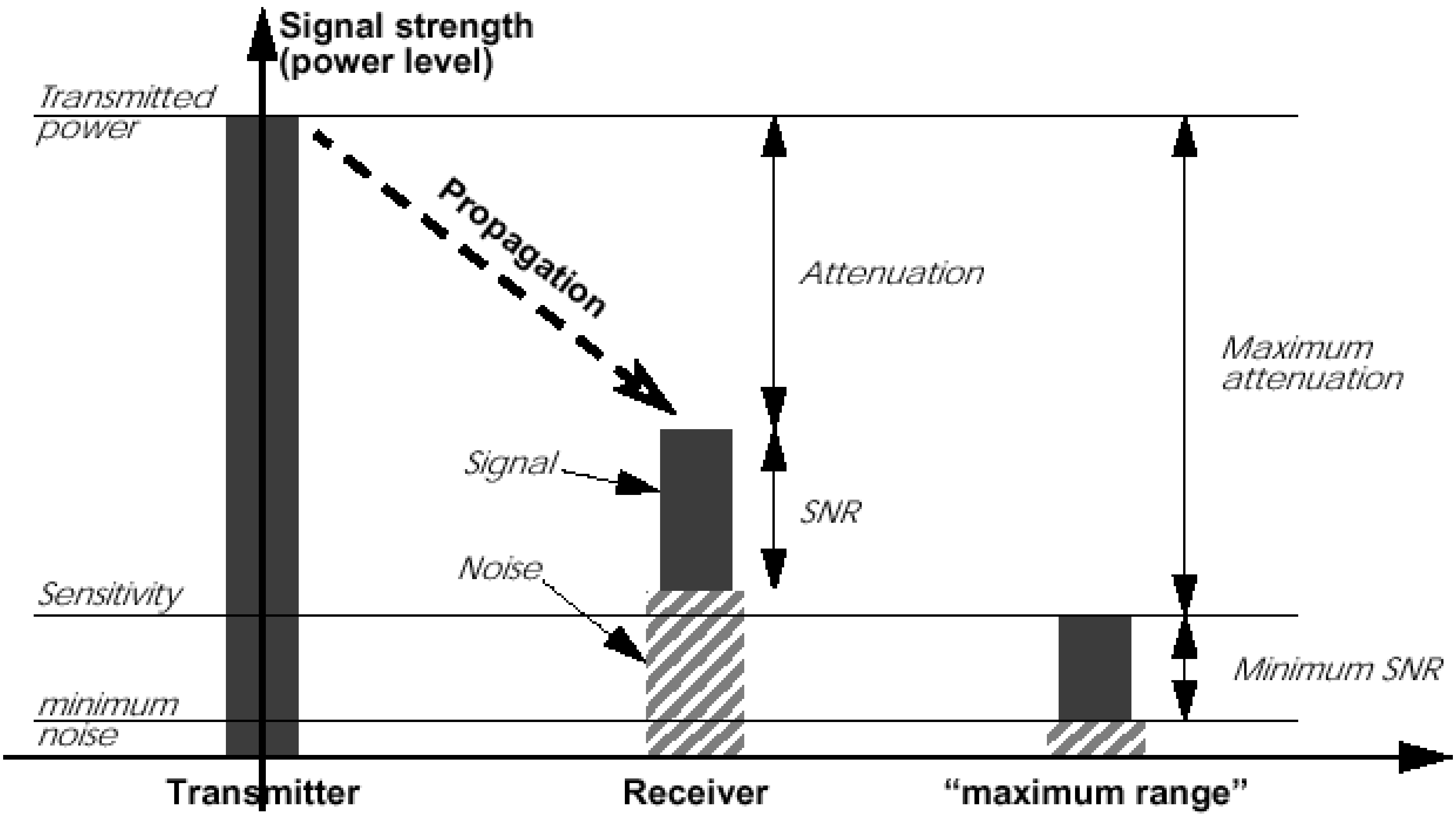
Typical Picture



Attenuation

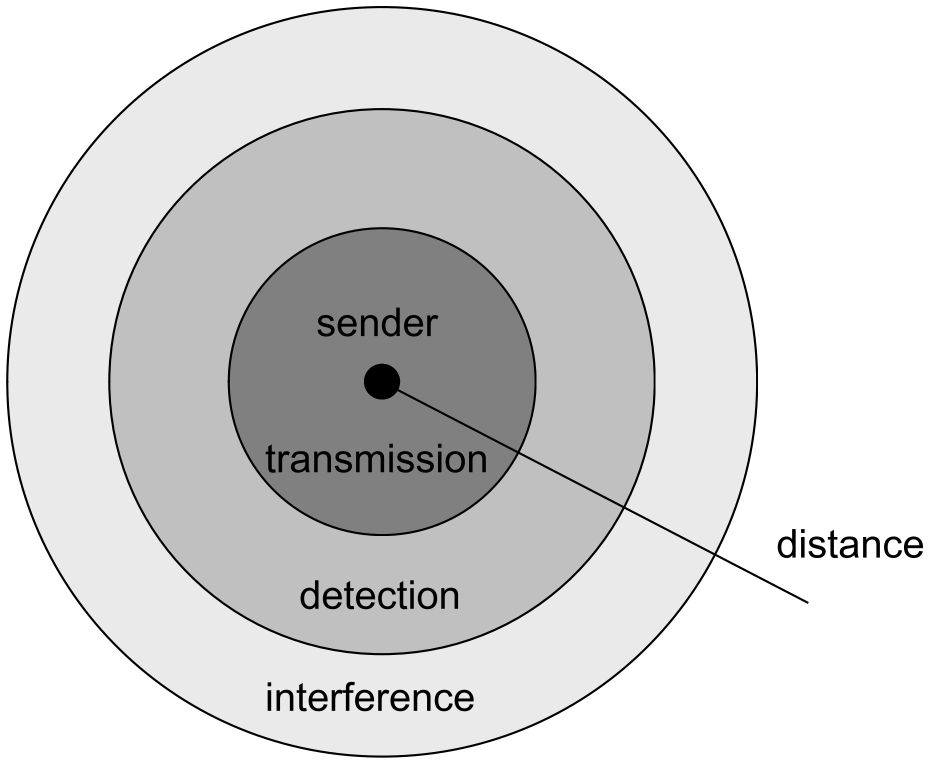
- Strength of signal falls off with distance over transmission medium
- Attenuation factors for unguided media:
  - Received signal must have sufficient strength so that circuitry in the receiver can interpret the signal
  - Signal must maintain a level sufficiently higher than noise to be received without error
  - Attenuation is greater at higher frequencies, causing distortion
- Approach: amplifiers that strengthen higher frequencies

Attenuation: Propagation & Range



Signal Propagation Ranges

- Transmission range
  - communication possible
  - low error rate
- Detection range
  - detection of the signal possible
  - no communication possible
- Interference range
  - signal may not be detected
  - signal adds to the background noise



Signal, Noise, and Interference

- Signal (S)
- Noise (N)
  - Includes thermal noise and background radiation
- Interference (I)
  - Signals from other transmitting sources
- SINR = S/(N+I) (sometimes also denoted as SNR)
- Large SINR = easier to extract signal from noise

dB and Power conversion

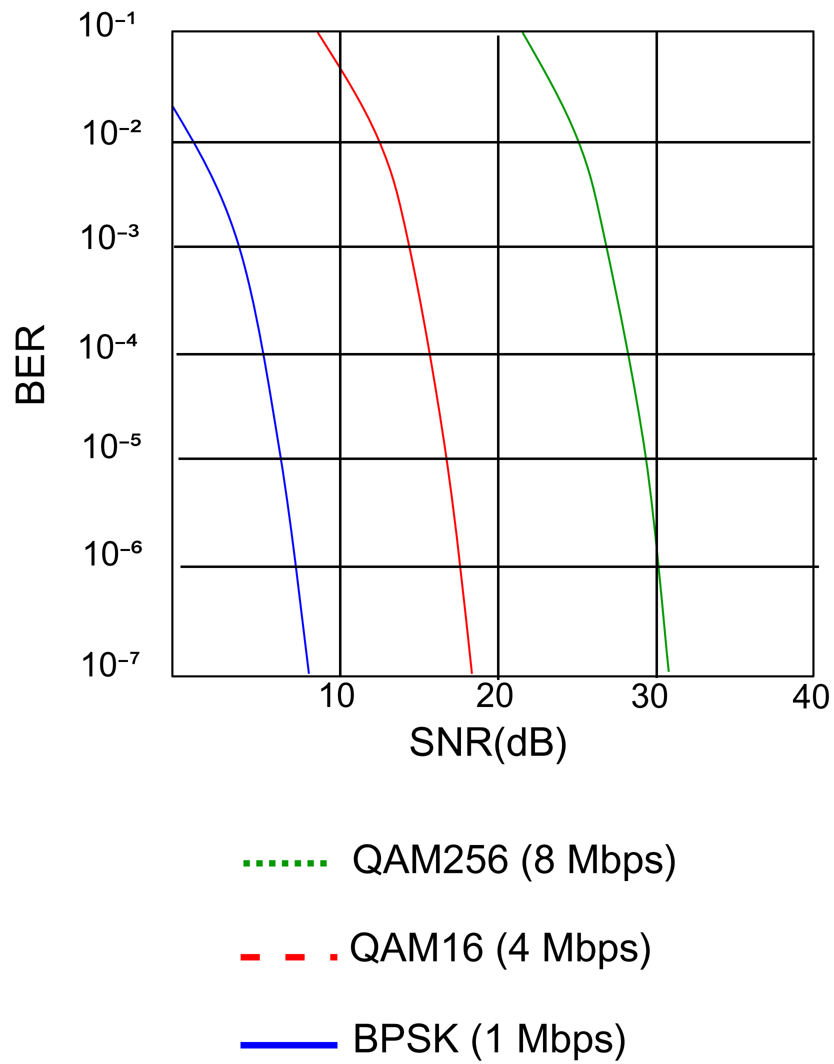
- dB
  - Denote the difference between two power levels
  - (P2/P1)[dB] = 10 * log10 (P2/P1)
  - P2/P1 = 10^(A/10)
  - Example 1: P2 = 10 P1, P2/P1=10dB
  - Example 2: P2/P1 = 33dB, P2 = 2000 P1

10dB: factor of 10
3dB: factor of 2

- dBm and dBW
  - Denote the power level relative to 1 mW or 1 W
  - P[dBm] = 10*log10(P/1mW)
  - P[dBW] = 10*log10(P/1W)
  - Example: P = 0.001mW = -30dBm, P = 100W = 20dBW

Wireless Link Characteristics

- SNR: signal-to-noise ratio
  - larger SNR – easier to extract signal from noise
- *SNR versus BER tradeoffs*
  - *given physical layer:* increase power -> increase SNR->decrease BER
  - *given SNR:* choose physical layer that meets BER requirement, giving highest throughput

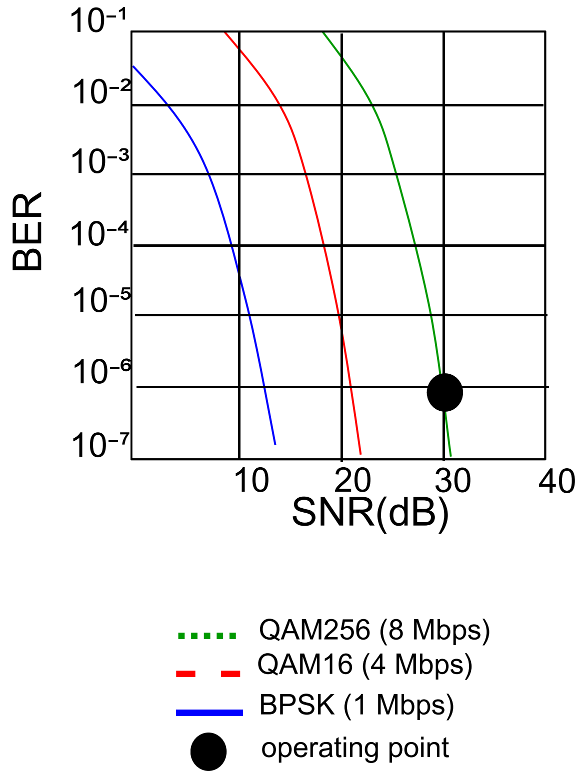


Rate Adaptation

- base station, mobile dynamically change transmission rate (physical layer modulation technique) as mobile moves, SNR varies

1. SNR decreases, BER increase as node moves away from base station

2. When BER becomes too high, switch to lower transmission rate but with lower BER



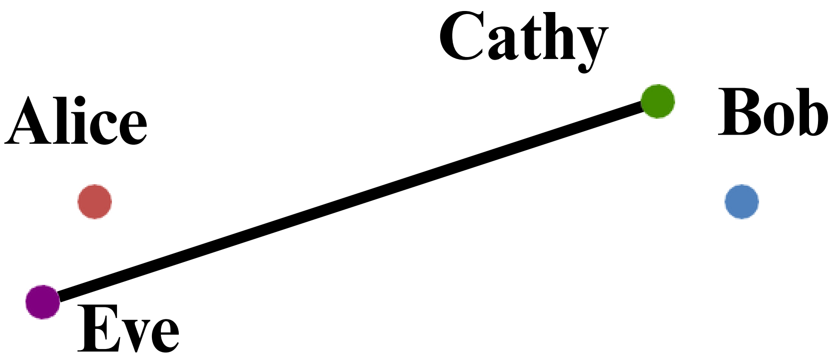
Understanding Wireless Communication: Barriers

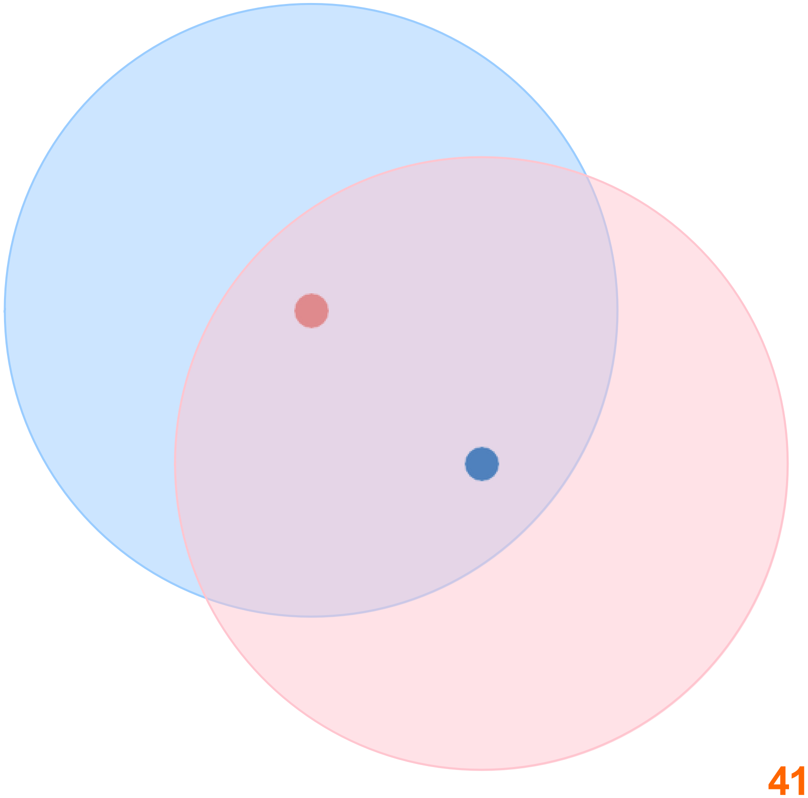
Wireless is less reliable

- In wired networks, link **bit error rate** is **10^{-12} and less**
- Wireless networks are **far from that target**
  – Bit error rates of **10^{-6} and above** are common!
- ***Why?***

Wireless is a shared medium

- **Wired networks:**
  Alice and Bob's conversation is **independent of** Cathy and Eve's conversation





- **Wireless networks:**
  Close by **wireless** conversations **share the same wireless medium**

Wireless Links: High Bit Error Rate

"Wireless communication often faces disruptions from the environment, causing more errors in the received data, known as a high Bit Error Rate (BER)."

Factors Contributing to High BER::

1. Signal Attenuation:
   - Signal gets weaker with distance or obstacles, making it harder to receive clearly.
2. Noise and Interference:
   - Other electronic signals mix with the intended one, causing data errors.
3. Multipath Fading & Environmental Obstructions:
   - Reflected signals arrive at different times, distorting the original message.
4. Doppler Shift (Mobility):
   - Movement causes frequency changes, making signal decoding difficult.
5. Broadcast Limitations at Wireless Links

Dealing With Bit Errors

- Wireless vs. wired links
  - Wired: most loss is due to congestion
  - Wireless: higher, time-varying bit-error ate
- Dealing with high bit-error rates
  - Sender could increase transmission power
    - Requires more energy (bad for battery-powered hosts)
    - Creates more interference with other senders
  - Stronger error detection and recovery
    - More powerful error detection codes
    - Link-layer retransmission of corrupted frames

1. Signal Attenuation

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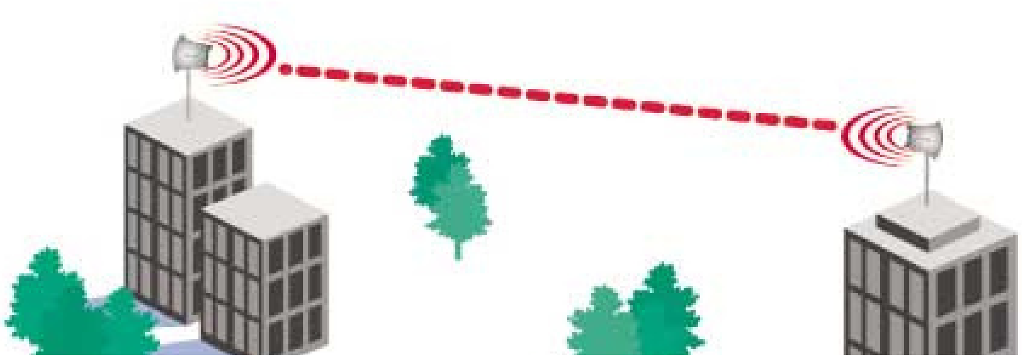
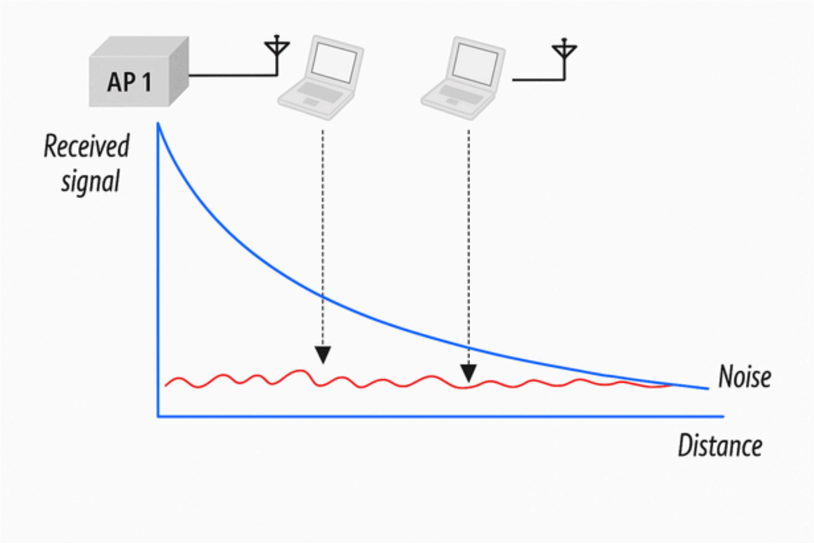
- Signal Attenuation:: Decreasing signal strength
  - Disperses as it travels greater distance
  - Attenuates as it passes through matter



1.Signal Attenuation:: High BER

- Decreasing signal strength
  - Disperses as it travels greater distance
  - Attenuates as it passes through matter
    - Metal particles present in air
    - Noice and interference

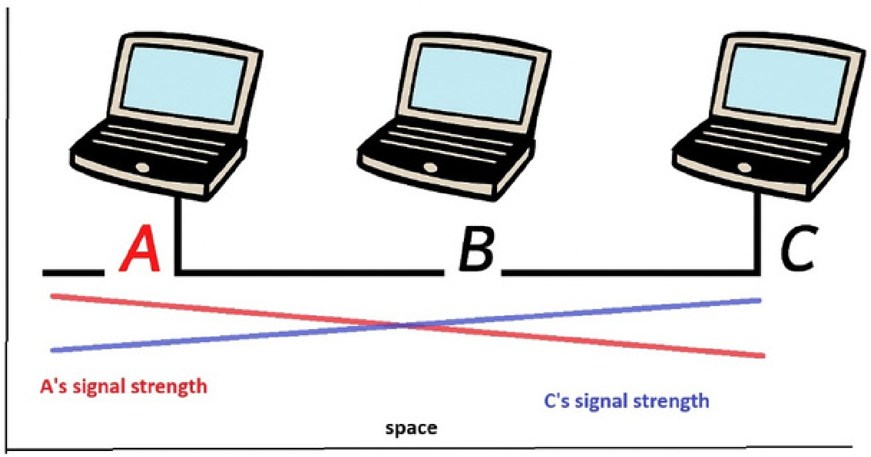


Lower signal-to-noise ratio (SNR) increases the probability of bit errors

1.Signal Attenuation

Received Signal Strength:: Wired vs Wireless

Received Signal Strength in
Wired Networks



Received Signal Strength in
Wireless Networks

A

B

C

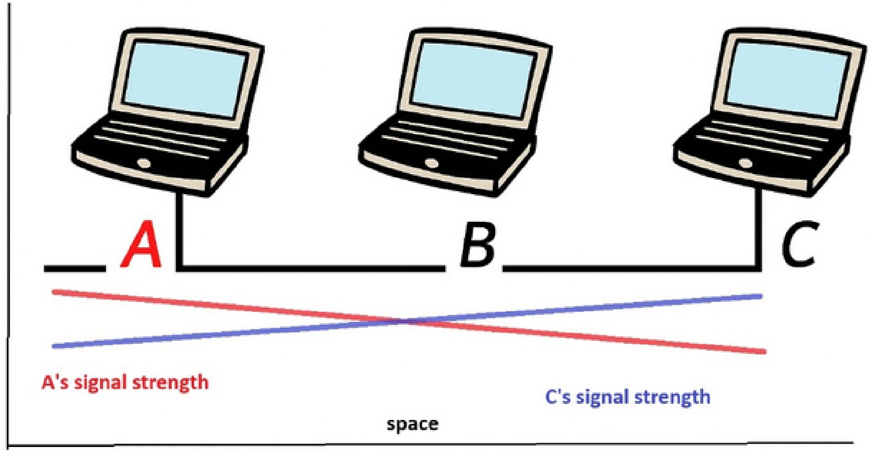
A's
signal
strengt
h

C's
signal
strengt
h

spac
e

1.Signal Attenuation

Received Signal Strength in Wired Networks



Reason: Minimal Signal Loss in Wired

- Signals travel through **guided media** like copper or fiber.
- These media **shield signals** from external interference and **minimize attenuation**.
- Losses due to resistance or cable imperfections are **very small** over short-to-moderate distances.

1.Signal Attenuation

Received Signal Strength in Wireless Networks

A

B

C

A's signal strength

C's signal strength

space

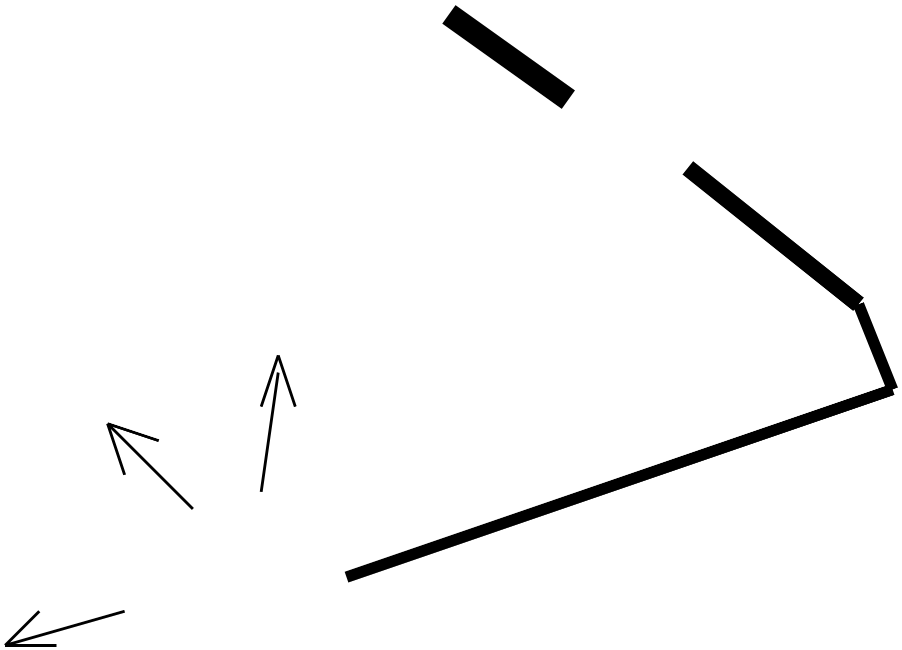
Reason: Higher Signal Loss in Wireless

In wireless networks, RSS **drops rapidly** with distance due to:

- **Path loss**
- **Obstacles**
- **Noise and interference**

1.Signal Attenuation

The higher frequencies will usually encounter more “loss” in “real world” situations



2. Noise and Interference

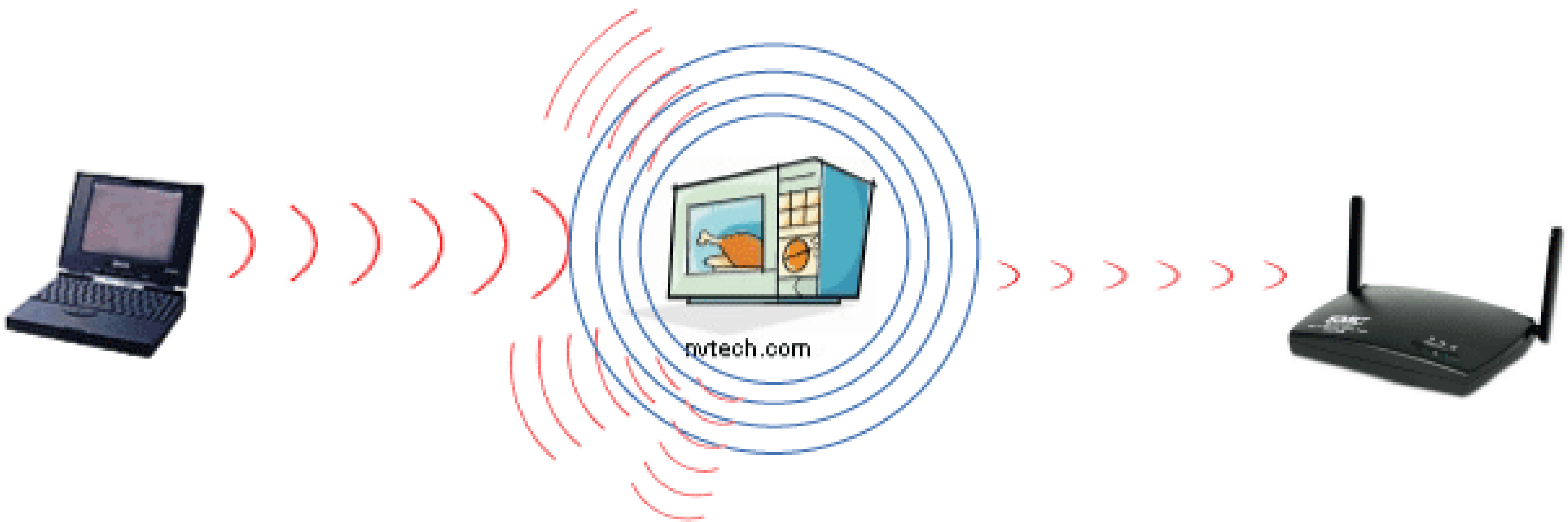
2. Noise and Interference

Categories of Noise

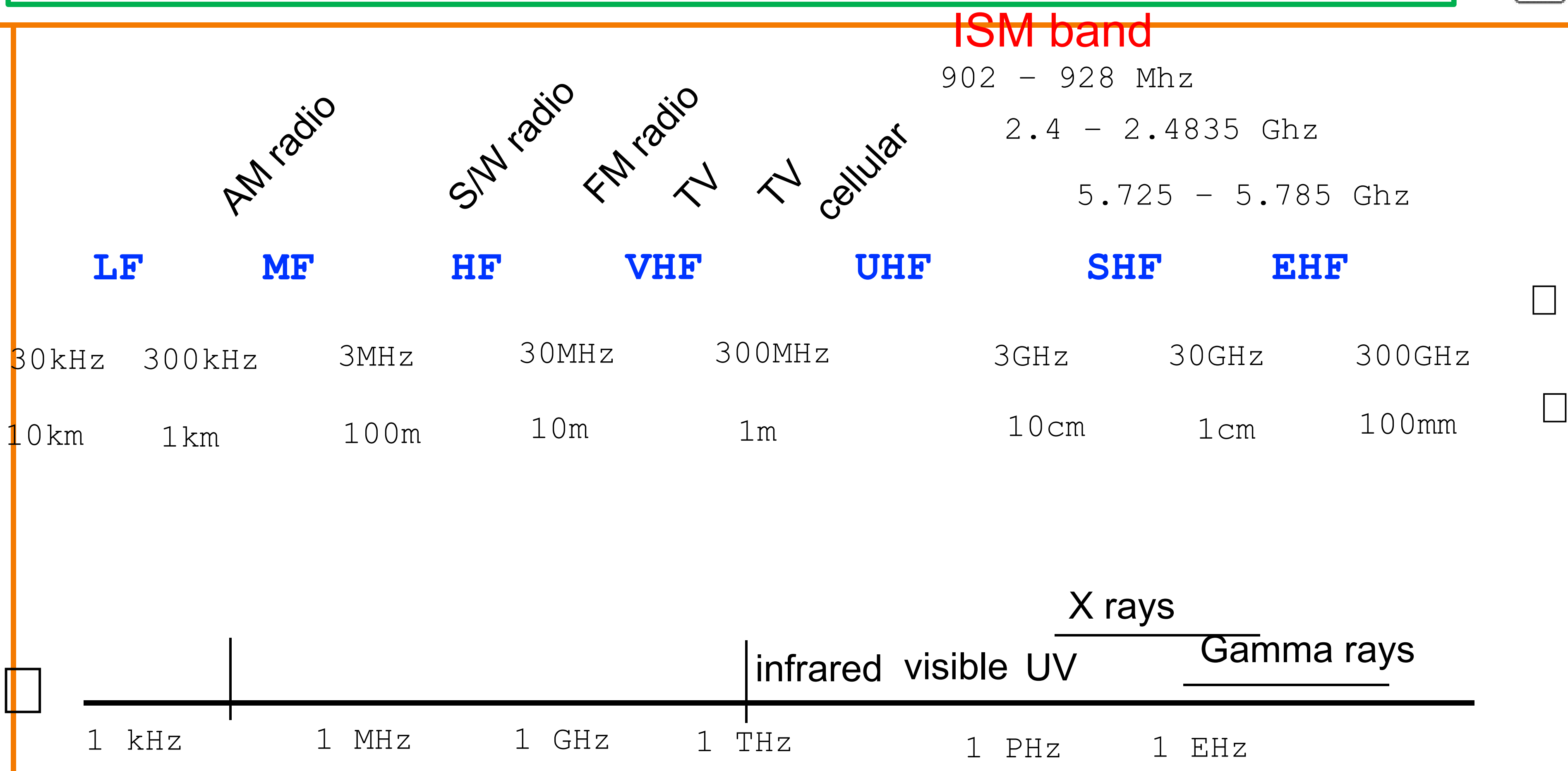
- Thermal Noise
  - Caused by the random motion of electrons in a conductor.
  - Present in all electronic devices and transmission media.
  - Increases with temperature; affects all frequencies uniformly.
- Intermodulation noise
  - Occurs when **signals at different frequencies mix** and produce additional unwanted signals.
- Crosstalk
  - Unwanted **signal leakage** between adjacent cables or channels.
- Impulse Noise
  - Sudden, short-duration high-energy spikes.
  - Caused by lightning, switching equipment, or power line disturbances.
  - Particularly harmful to digital communication; can corrupt multiple bits at once.

2. Noise and Interference

- Interference from other sources
  - Radio sources in same frequency band
  - E.g., 2.4 GHz wireless phone interferes with 802.11b wireless LAN
  - Electromagnetic noise (e.g., microwave oven)



EM Spectrum



Propagation characteristics are different in each frequency band

EM Spectrum:: ISM Band

ISM Band (Industrial, Scientific, and Medical Band)

Key Features

- License-free usage (no regulatory approval needed, with power limits).
- Common in Wi-Fi, Bluetooth, RFID, microwave ovens, and cordless phones.
- Operates under power and interference constraints to coexist with other users.

This can cause **interference**, especially between Wi-Fi and microwave ovens.

Modern Wi-Fi devices often switch to 5 GHz or 6 GHz to avoid congestion.

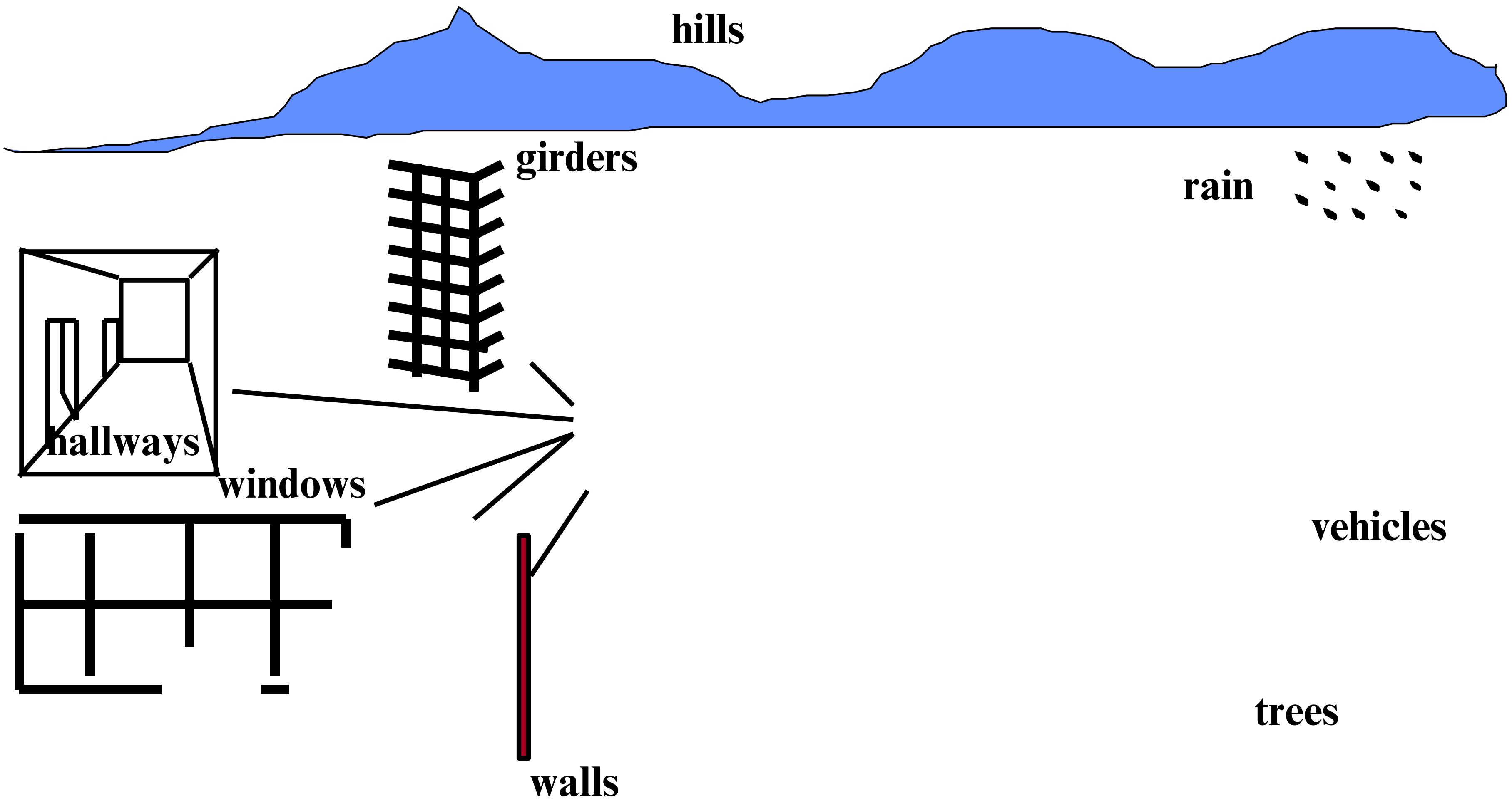
2.Noise and Interference:: High BER

How Noise & Interference Increase Bit Error Rate (BER)

- Noise (thermal, impulse) distorts the signal, making it hard to detect correct bits.
- Interference from nearby devices overlaps signals, causing bit confusion.
- Low SNR reduces the receiver's ability to distinguish between 0s and 1s.

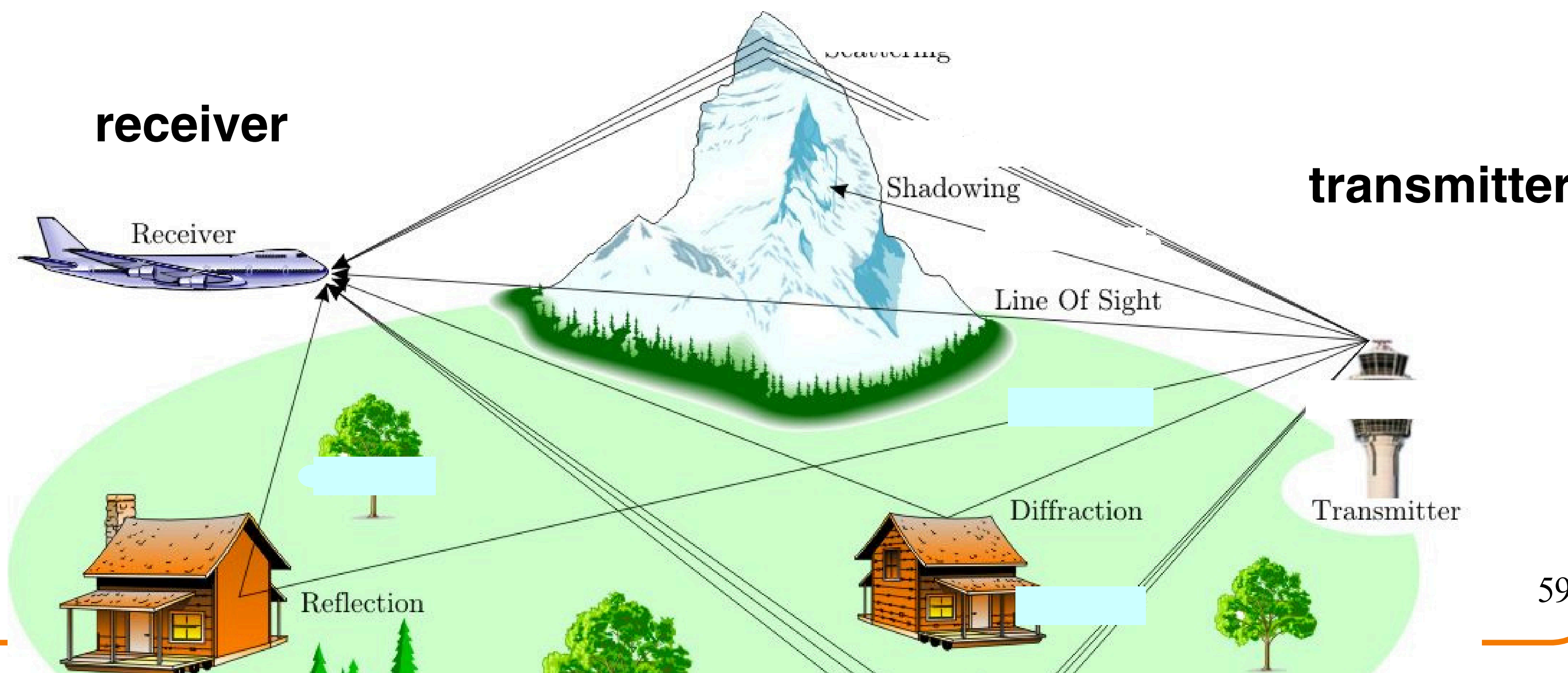
3. Multipath Fading & Environmental Obstructions

The Cluttered World of Radio Waves



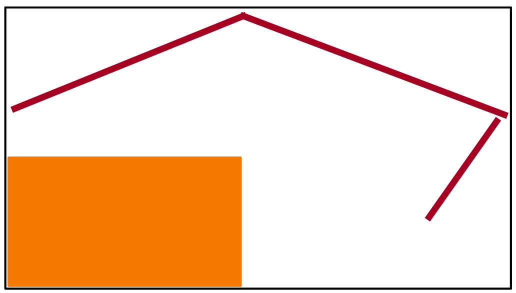
3.Multipath Fading & Environmental Obstructions: High BER

- Multi-path propagation
  - Electromagnetic waves reflect off objects
  - Taking many paths of different lengths
  - Causing blurring of signal at the receiver
  - **Multipath fading** causes signal distortion and delays, increasing errors (BER).



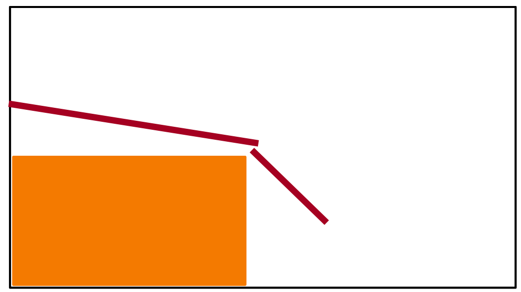
3.Multipath Fading & Environmental Obstructions: High BER

Three basic propagation mechanisms



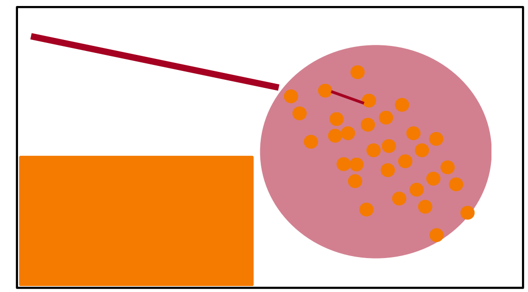
Reflection

$\lambda << D$



Diffraction

$\lambda \approx D$



Scattering

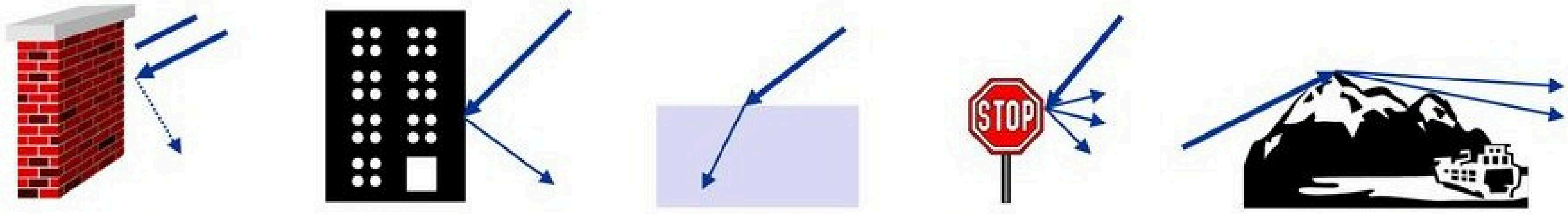
$\lambda >> D$

- Propagation effects depend on not only on the specific portion of spectrum used for transmission, but also on the bandwidth (or spectral occupancy) of the signal being transmitted

- Spatial separation of Tx-Rx

3.Multipath Fading & Environmental Obstructions: High BER

- Propagation in free space is always like light (straight line).
- Receiving power proportional to **1/d²** in vacuum – much more in real environments (**d** = distance between sender and receiver)
- Receiving power additionally influenced by
  - fading (frequency dependent)
  - Shadowing (blocking)
  - reflection at large obstacles
  - refraction depending on the density of a medium
  - scattering at small obstacles
  - diffraction at edges



shadowing reflection refraction scattering diffraction

4. Doppler Shift (Mobility):

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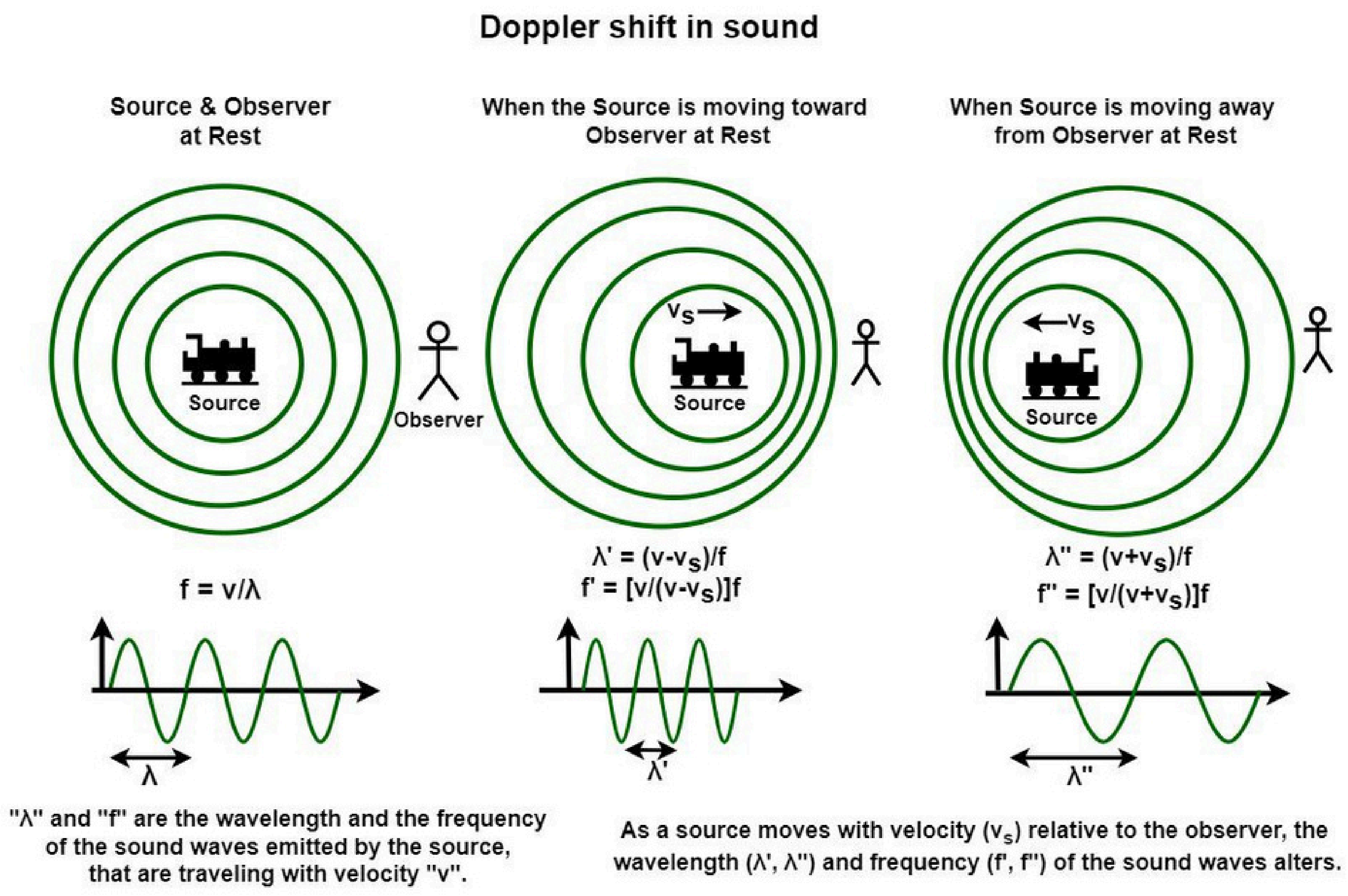


4.Doppler Shift (Mobility): High BER

- **Doppler Shift** causes **frequency changes** in the received signal due to relative motion between transmitter and receiver.
- This leads to **carrier frequency offset** and **loss of synchronization** at the receiver.
- It also increases **channel variation rate**, making the channel **time-varying and unpredictable**.
- As a result, symbols may be **misinterpreted**, especially in high-speed mobility scenarios.

Impact
→ Distorted signal → Synchronization errors → **Higher BER**

5. Broadcast Limitations at Wireless Links

5.Wireless Links: Broadcast Limitations

- Wired broadcast links
  - E.g., Ethernet bridging, in wired LANs
  - All nodes receive transmissions from all other nodes
- Wireless broadcast: hidden terminal problem

C

B

A

- **A and B hear each other**
- **B and C hear each other**
- **But, A and C do not**

So, A and C are

5.Wireless Links: Broadcast Limitations

- Wired broadcast links
  - E.g., Ethernet bridging, in wired LANs
  - All nodes receive transmissions from all other nodes
- Wireless broadcast: fading over distance

A B C

A's signal strength

C's signal strength

space

- A and B hear each other
- B and C hear each other
- But, A and C do not

So, A and C are

5.Wireless Links: Broadcast Limitations

Why is this a problem in wireless networks?

C

B

A

A

B

C

A's signal strength

C's signal strength

space

5.Wireless Links: Broadcast Limitations

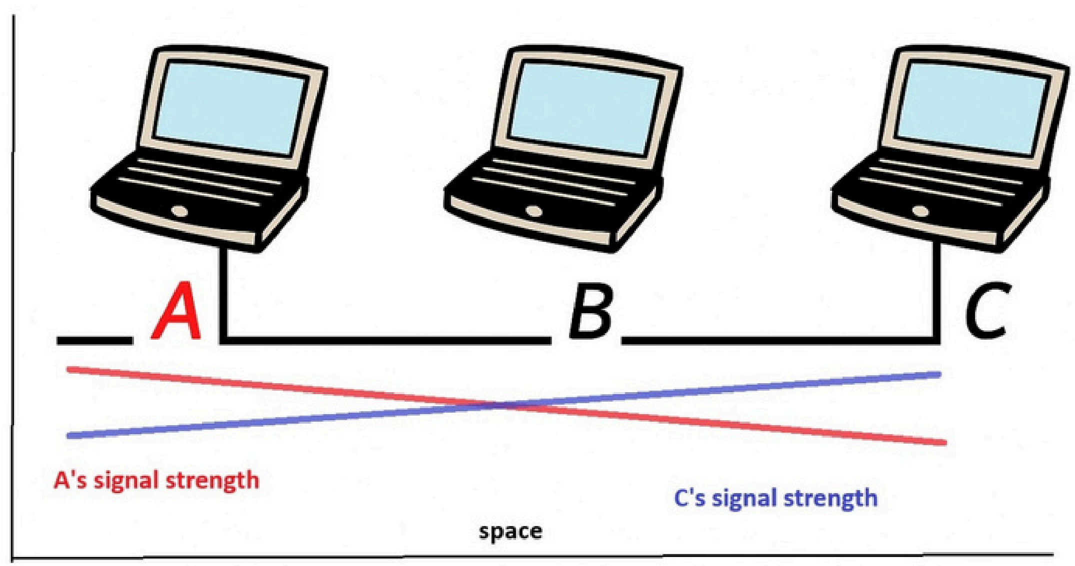
Why is it NOT a problem in wired networks?

- In wired networks, like Ethernet:
  - All nodes are also connected to a shared medium (the wire).
  - Every device can “hear” all transmissions with acceptable received Signal Strength, so they avoid sending at the same time.
  - (CSMA/CD) Collision Detection works well because all signals are detectable.

C

B

A



5.Wireless Links: Broadcast Limitations:: High BER

Impact

→ Undetected collisions at receiver → Corrupted data → **Increased BER**

A

B

C

A's signal strength

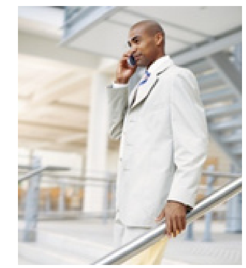
C's signal strength

space

Understanding Wireless Communication: Needs, Barriers, and Design Adaptations

Understanding Wireless Communication: Needs

Examples of Communication Requirements



Cell phones:

bit error probability: < 0.1 %
delay: < 100 ms
data rate: 10-15 kbit/s

E-mailing:

bit error probability: $< 10^{-9}$
delay: < 1 day
data rate: irrelevant

Mobile broadband:

bit error probability: $< 10^{-5}$
delay: < seconds
data rate: < 80 Mbit/s

Multimedia Requirements

	Voice	Data	Video
Delay	<100ms	-	<100ms
Packet Loss	<1%	0	<1%
BER	10^{-3}	10^{-6}	10^{-6}
Data Rate	8-32 Kbps	1-100 Mbps	1-20 Mbps
Traffic	Continuous	Bursty	Continuous

One-size-fits-all protocols and design do not work well

Wired networks use this approach, with poor results

Understanding Wireless Communication: Design Adaptations

Possible Design Goals

- Maximize the transmission rate
- Minimize the probability of error
- Minimize the required power
- Minimize the required system bandwidth
- Maximize system utilization
- Minimize system complexity
- Minimize communication delay

Some of these goals are in conflict with each other!

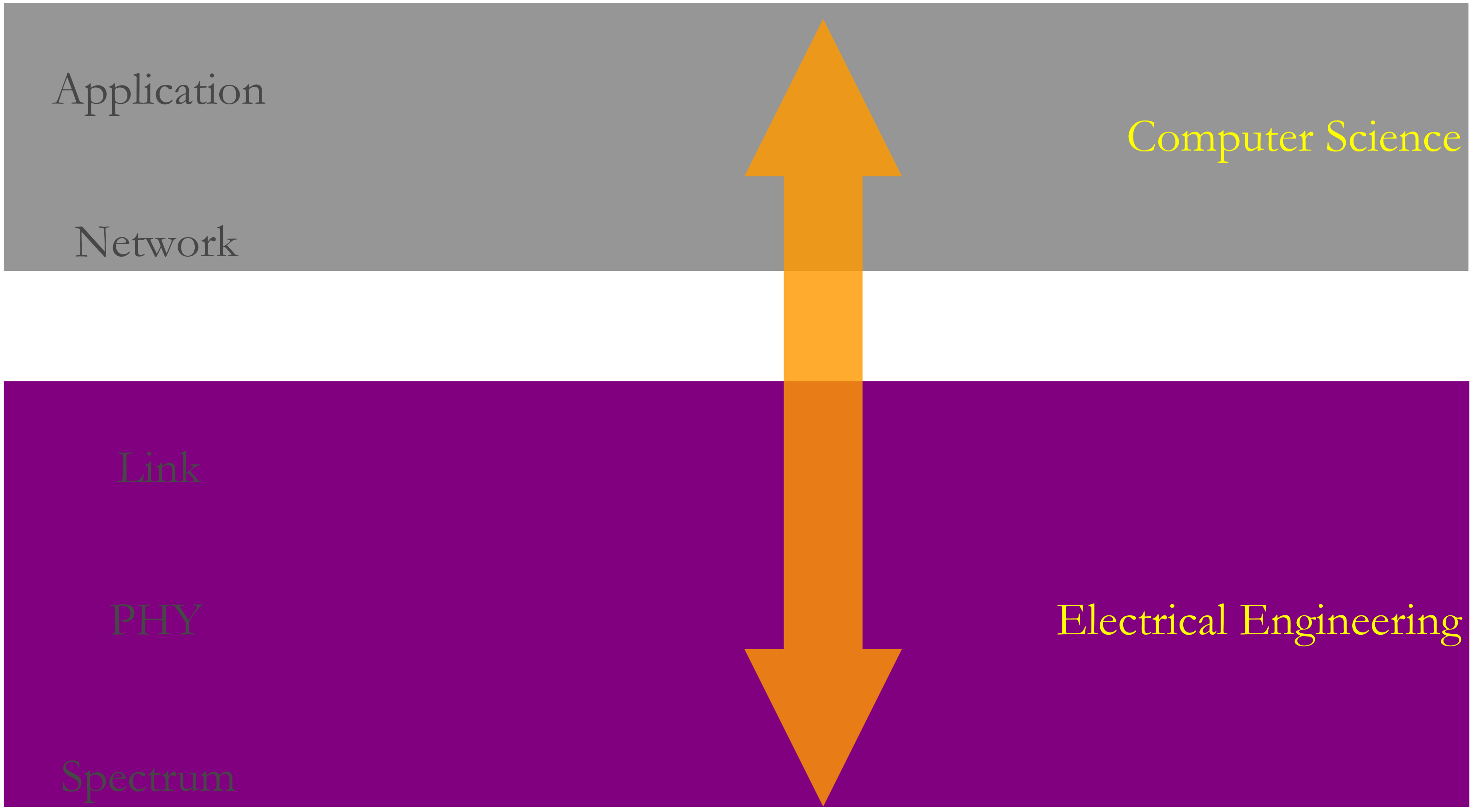
Evolution of Current Systems

- Recent past
  - 2G Cellular: ~30-70 Kbps.
  - WLANs: ~10 Mbps.
  - 3G Cellular: ~300 Kbps.
  - WLANs: ~70 Mbps.
- Technology Enhancements
  - Hardware: Better batteries. Better circuits/processors.
  - Link: Antennas, modulation, coding, adaptivity, DSP, BW.
  - Network: Dynamic resource allocation. Mobility support.
  - Application: Soft and adaptive QoS.

Design Challenges

- Wireless channels are a difficult and capacity-limited broadcast communications medium
- Traffic patterns, user locations, and network conditions are constantly changing
- Applications are heterogeneous with hard constraints that must be met by the network
- Energy and delay constraints change design principles across all layers of the protocol stack

OSI Model


Application
Network
Computer Science
Link
PHY
Spectrum
Electrical Engineering

Crosslayer Design

- Hardware
- Link
- Access
- Network
- Application

Delay Constraints
Rate Constraints
Energy Constraints

Adapt across design layers
Reduce uncertainty through scheduling
Provide robustness via diversity